

TidyTuesday Week 37

Pranay Gundam

Thursday 10th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to another week of the world's most rigorously pointless science experiment, where we let a random number generator introduce two total strangers from the FRED database and then act shocked when they have chemistry. As always, the disclaimer up top: nothing here means anything. These pairings are drawn at random, there is no theory, no hypothesis, and absolutely no reason to believe any of these series have ever influenced one another. Spurious correlation isn't the bug here – it's the entire point. Now let's rummage through the wreckage.

Monday kicked things off with a genuine crowd-pleaser: the Discussion About Pandemics Index for Lithuania versus the U.S. government's Borrowing from the Public. Naturally. Because when Lithuanians start chatting about pandemics, who among us doesn't immediately think of American federal financing? The raw regression flagged a significant relationship, and – brace yourself – it actually survived detrending, holding an R-squared around 0.17 with a p-value that stayed comfortably tiny. So this is one of the rare ones that doesn't instantly collapse into "they both drifted upward, case closed." Is there a real mechanism linking Baltic pandemic chit-chat to Treasury borrowing? Absolutely not. But the numbers held up, so I'm legally obligated to raise an eyebrow before moving on with my life.

Tuesday and Wednesday were more of the same beautiful nonsense. First, U.S. coal-based CO2 emissions squared off against a price index for medical services for congenital anomalies, and this one was almost suspiciously strong – a raw R-squared over 0.6 that not only survived detrending but actually got a punchier coefficient once the trends were stripped out. Two totally unrelated series clinging to each other like that is exactly the kind of statistical drama this project lives for, so enjoy it, and then remind yourself it means nothing. Wednesday's matchup, the Market Value of Gross Federal Debt against business equipment leases owned by finance companies, was the more familiar trend-mirage-adjacent story: a raw relationship that flipped its sign and stayed significant after detrending, which is honestly just the two series being messy in ways I refuse to interpret further.

Then Thursday arrived to restore balance to the universe. Construction-sector quits versus the work-from-home rate for women produced a raw regression so limp it had a p-value of 0.86 and an R-squared of basically zero – the statistical equivalent of two people making eye contact and immediately looking away. But plot twist: once detrended, it suddenly coughed up an R-squared near 0.3 and a very significant coefficient, meaning the "nothing to see here" raw result was hiding an actual co-movement once the trends were removed. Which is the exact opposite of the usual dunk, and a nice reminder that the raw numbers lie in both directions. Anyway, that's

the week: several relationships that stubbornly refused to die under detrending, none of which you should mention in polite company as though they were real. See you next week for more randomly assembled economic soulmates.

2 Date: 2026-09-07

Series ID: WPDILTU

This series is titled Discussion About Pandemics Index for Lithuania and has a frequency of Quarterly. The units are Index and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1996-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: MTSMFBP133FMS

This series is titled Means of Financing: Borrowing from the Public and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2010-01-01 and the observation end date is 2026-07-01. The popularity of this series is 14.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MTSMFBP133FMS	R-squared:	0.199			
Model:	OLS	Adj. R-squared:	0.186			
Method:	Least Squares	F-statistic:	15.87			
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000176			
Time:	16:43:46	Log-Likelihood:	-893.46			
No. Observations:	66	AIC:	1791.			
Df Residuals:	64	BIC:	1795.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	4.886e+04	2.46e+04	1.985	0.051	-321.644	9.8e+04
value_fred_WPDILTU	1205.6312	302.613	3.984	0.000	601.092	1810.170
Omnibus:	51.230	Durbin-Watson:	2.093			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	275.412			
Skew:	2.139	Prob(JB):	1.57e-60			
Kurtosis:	12.047	Cond. No.	87.5			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

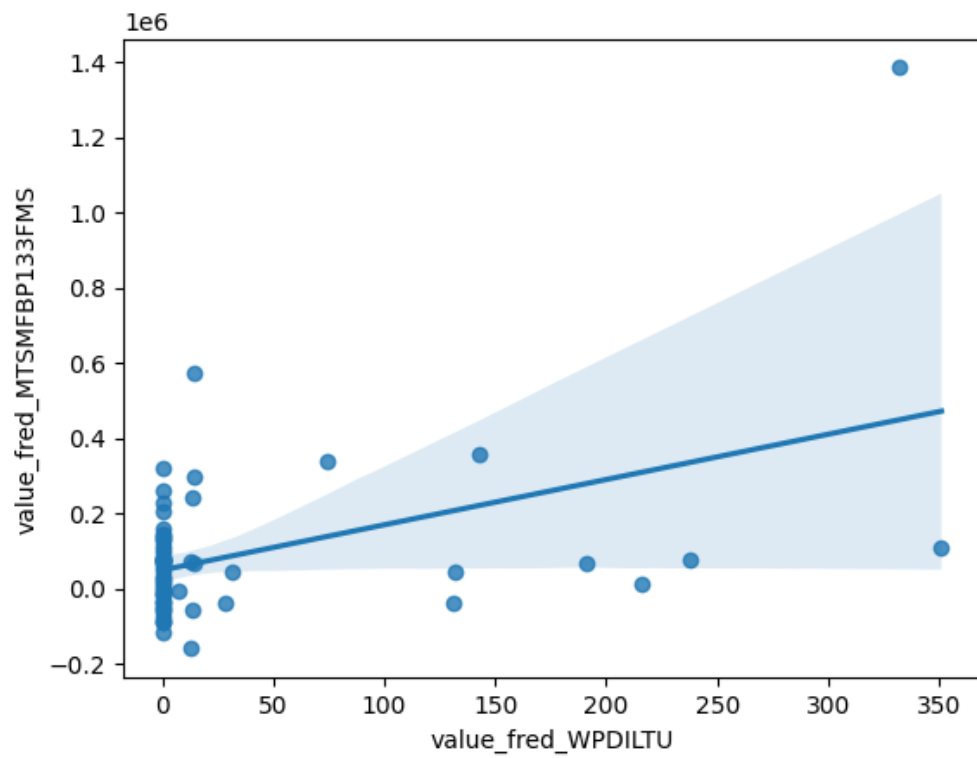


Figure 1: Raw Regression Plot for 2026-09-07

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MTSMFBP133FMS_detrended	R-squared:	0.174
Model:	OLS	Adj. R-squared:	0.161
Method:	Least Squares	F-statistic:	13.48
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000493
Time:	16:43:46	Log-Likelihood:	-893.06
No. Observations:	66	AIC:	1790.
Df Residuals:	64	BIC:	1794.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t 	[0.025	0.975]
const	5.051e-09	2.28e+04	2.22e-13	1.000	-4.55e+04	4.55e+04
value_fred_WPDILTU_detrended	1138.7087	310.105	3.672	0.000	519.204	1758.214

Omnibus:	52.012	Durbin-Watson:	2.129
Prob(Omnibus):	0.000	Jarque-Bera (JB):	302.597
Skew:	2.138	Prob(JB):	1.96e-66
Kurtosis:	12.579	Cond. No.	73.4

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

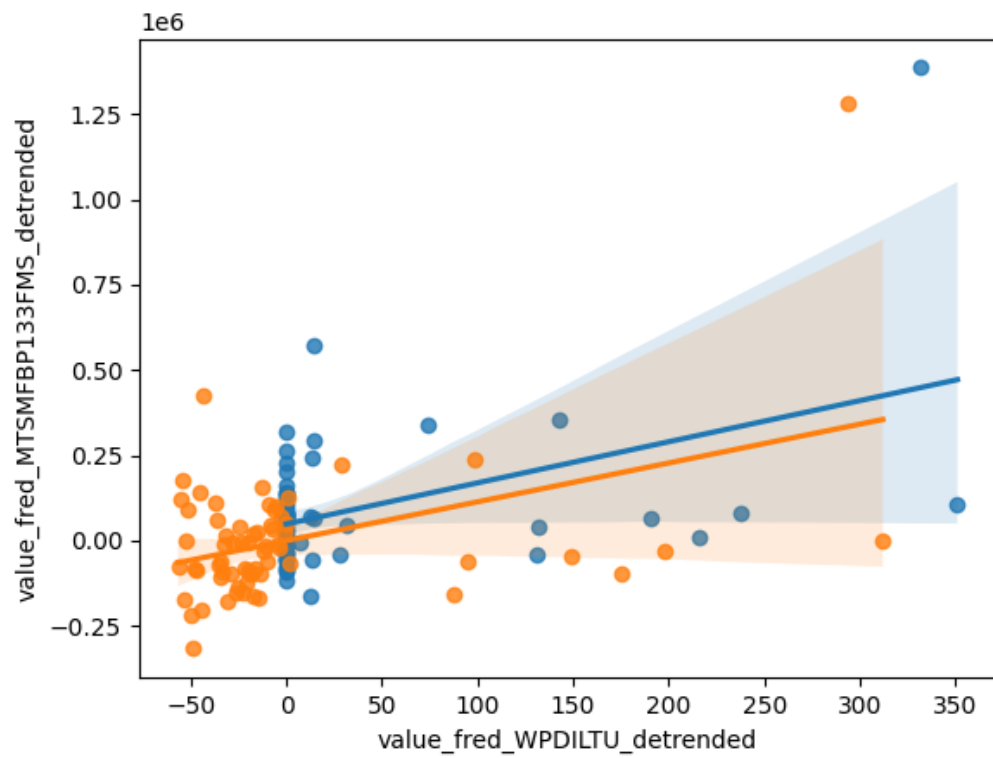


Figure 2: Detrended Regression Plot for 2026-09-07

3 Date: 2026-09-08

Series ID: EMISSCO2TOTVTTCOUSA

This series is titled Total Carbon Dioxide Emissions From All Sectors, Coal for United States and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1970-01-01 and the observation end date is 2021-01-01. The popularity of this series is 5.

Series ID: COGANOPIMEPS

This series is titled Medical Services Expenditures by Disease: Other: Congenital Anomalies Price Index, MEPS Account Basis and has a frequency of Annual. The units are Index 2009=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2020-01-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_COGANOPIMEPS	R-squared:	0.606
Model:	OLS	Adj. R-squared:	0.585
Method:	Least Squares	F-statistic:	29.22
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	3.24e-05
Time:	15:27:44	Log-Likelihood:	-111.42
No. Observations:	21	AIC:	226.8
Df Residuals:	19	BIC:	228.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-146.4574	51.777	-2.829	0.011	-254.827	-38.088
value_fred_EMISSCO2TOTVTTCOUSA	0.1533	0.028	5.406	0.000	0.094	0.213

Omnibus:	3.022	Durbin-Watson:	1.167
Prob(Omnibus):	0.221	Jarque-Bera (JB):	1.530
Skew:	0.625	Prob(JB):	0.465
Kurtosis:	3.429	Cond. No.	8.45e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.45e+03. This might indicate that there are strong multicollinearity or other numerical problems.

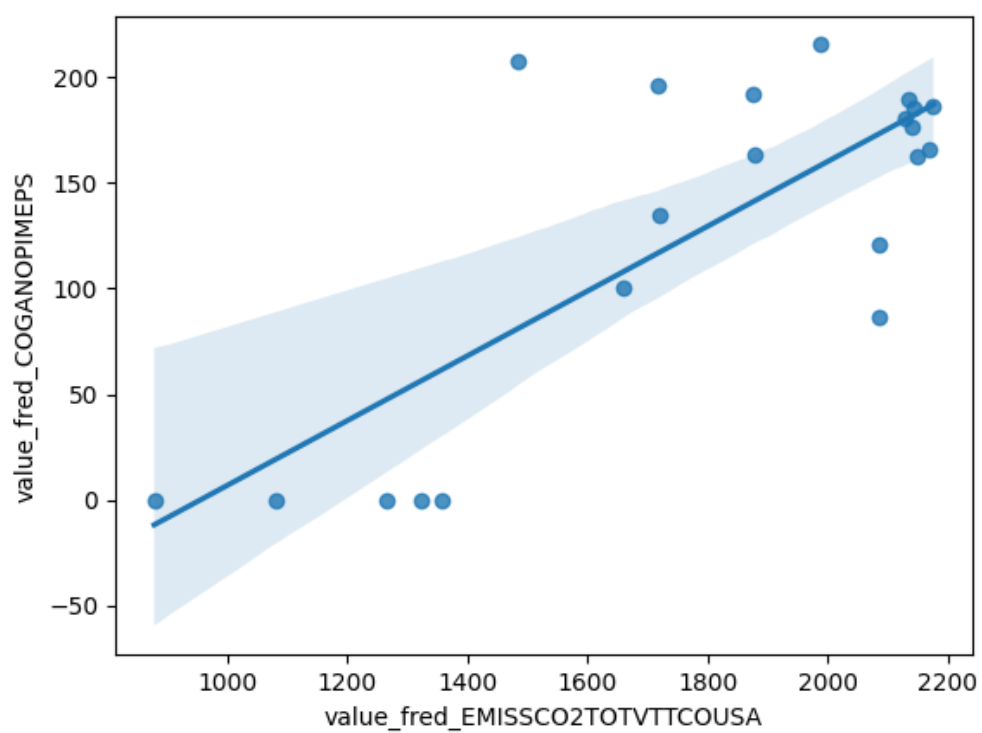


Figure 3: Raw Regression Plot for 2026-09-08

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_COGANOPIMEPS_detrended	R-squared:	0.559				
Model:	OLS	Adj. R-squared:	0.536				
Method:	Least Squares	F-statistic:	24.09				
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	9.76e-05				
Time:	15:27:44	Log-Likelihood:	-108.14				
No. Observations:	21	AIC:	220.3				
Df Residuals:	19	BIC:	222.4				
Df Model:	1						
Covariance Type:	nonrobust						
		coef	std err	t	P> t	[0.025	0.975
const		-6.734e-12	9.567	-7.04e-13	1.000	-20.023	20.023
value_fred_EMISSCO2TOTVTTCOUSA_detrended		0.3032	0.062	4.908	0.000	0.174	0.432
Omnibus:	4.692	Durbin-Watson:	1.541				
Prob(Omnibus):	0.096	Jarque-Bera (JB):	2.578				
Skew:	0.726	Prob(JB):	0.276				
Kurtosis:	3.915	Cond. No.	155.				

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

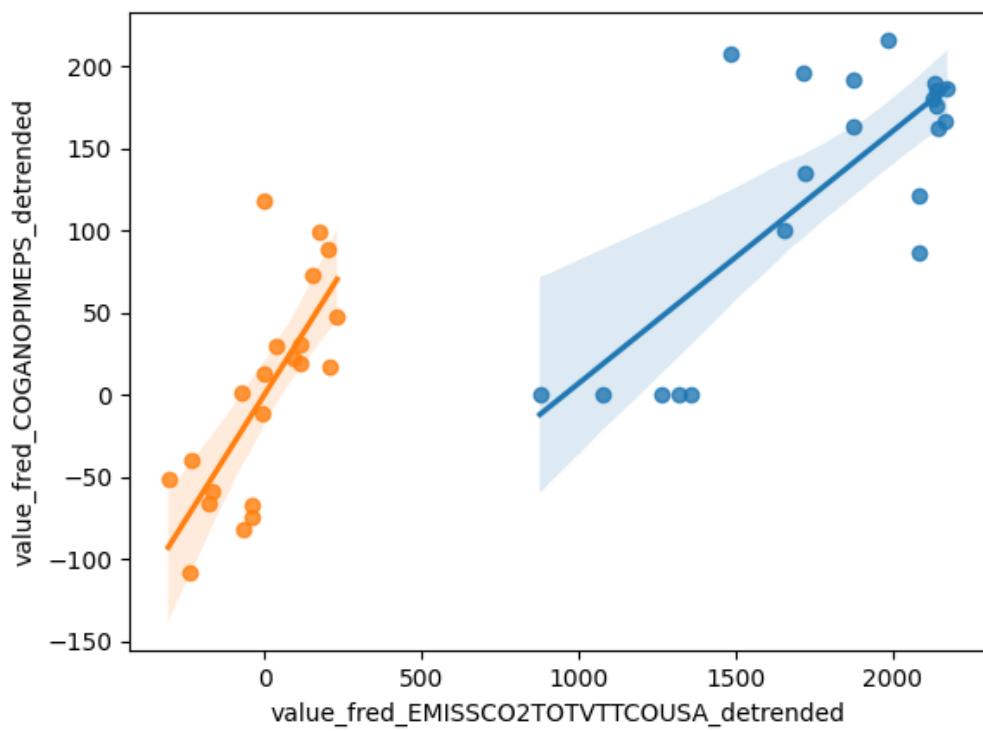


Figure 4: Detrended Regression Plot for 2026-09-08

4 Date: 2026-09-09

Series ID: MVGFD027MNFRBDAL

This series is titled Market Value of Gross Federal Debt and has a frequency of Monthly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1942-01-01 and the observation end date is 2026-07-01. The popularity of this series is 37.

Series ID: DTBTERNM

This series is titled Business Equipment Leases Owned and Securitized by Finance Companies, Level and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1990-06-01 and the observation end date is 2026-06-01. The popularity of this series is 1.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DTBTERNM	R-squared:	0.095
Model:	OLS	Adj. R-squared:	0.093
Method:	Least Squares	F-statistic:	45.33
Date:	Wed, 09 Sep 2026	Prob (F-statistic):	5.34e-11
Time:	15:24:33	Log-Likelihood:	-5318.4
No. Observations:	433	AIC:	1.064e+04
Df Residuals:	431	BIC:	1.065e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.633e+05	4472.957	36.517	0.000	1.55e+05	1.72e+05
value_fred_MVGFD027MNFRBDAL	-1.7290	0.257	-6.732	0.000	-2.234	-1.224

Omnibus:	2749.157	Durbin-Watson:	0.021
Prob(Omnibus):	0.000	Jarque-Bera (JB):	48.079
Skew:	0.054	Prob(JB):	3.63e-11
Kurtosis:	1.371	Cond. No.	3.10e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.1e+04. This might indicate that there are strong multicollinearity or other numerical problems.

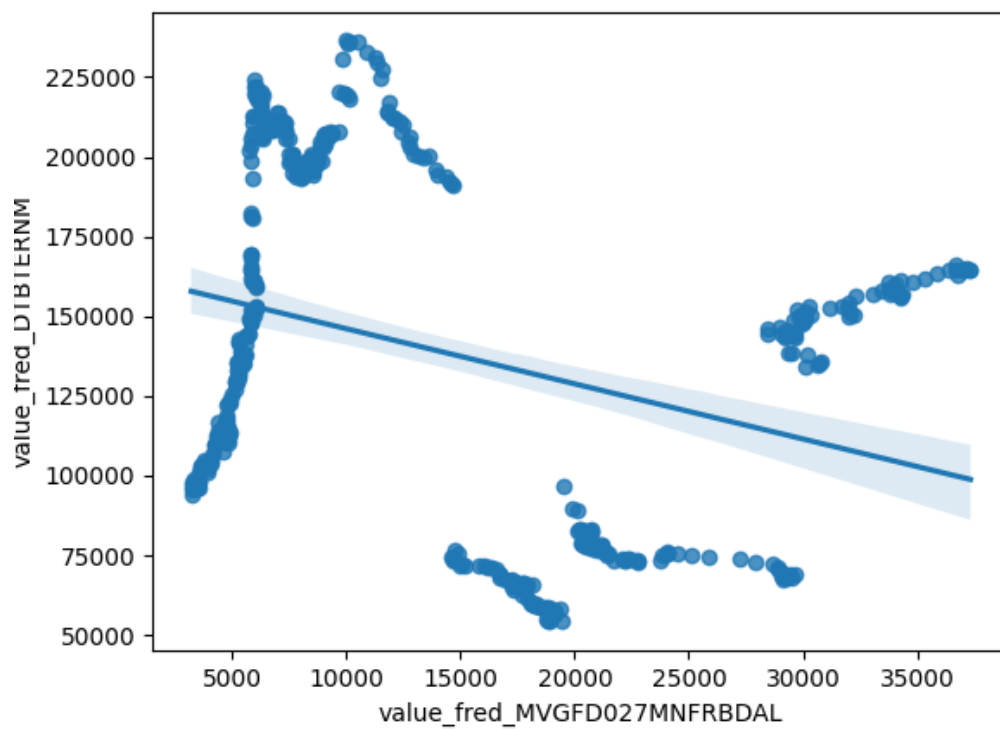


Figure 5: Raw Regression Plot for 2026-09-09

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DTBTERNM_detrended	R-squared:	0.116
Model:	OLS	Adj. R-squared:	0.114
Method:	Least Squares	F-statistic:	56.72
Date:	Wed, 09 Sep 2026	Prob (F-statistic):	2.97e-13
Time:	15:24:33	Log-Likelihood:	-5302.8
No. Observations:	433	AIC:	1.061e+04
Df Residuals:	431	BIC:	1.062e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975
const	9.044e-09	2427.545	3.73e-12	1.000	-4771.300	4771.300
value_fred_MVGFD027MNFRBDAL_detrended	-6.1487	0.816	-7.531	0.000	-7.753	-4.543

Omnibus:	698.313	Durbin-Watson:	0.023
Prob(Omnibus):	0.000	Jarque-Bera (JB):	33.214
Skew:	-0.174	Prob(JB):	6.13e-08
Kurtosis:	1.689	Cond. No.	2.97e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.97e+03. This might indicate that there are strong multicollinearity or other numerical problems.

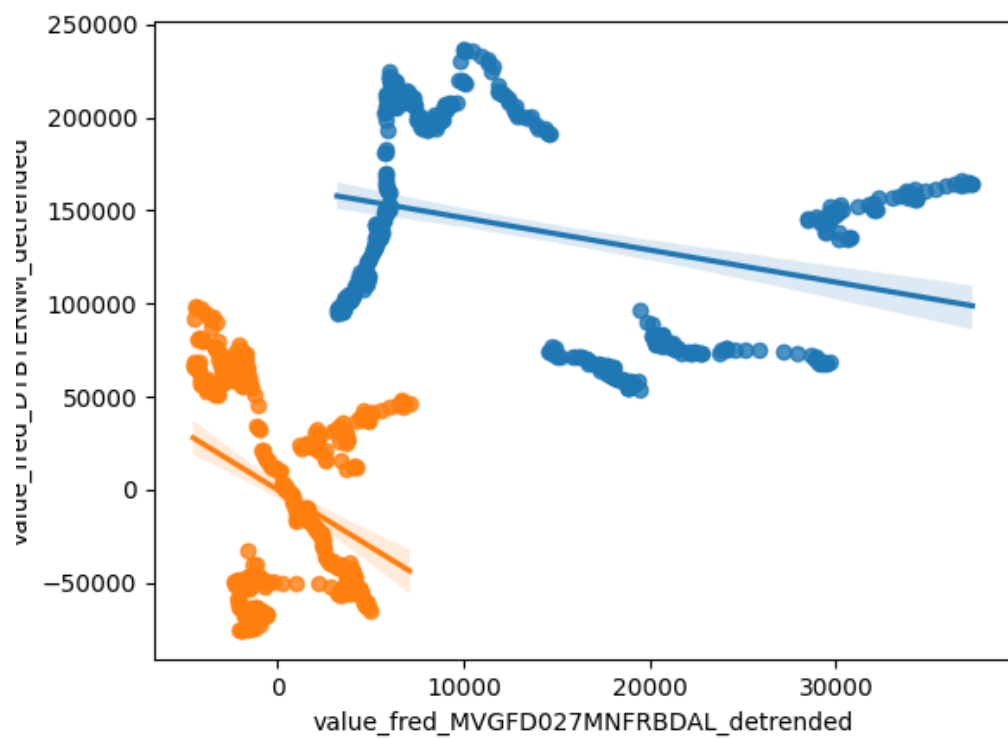


Figure 6: Detrended Regression Plot for 2026-09-09

5 Date: 2026-09-10

Series ID: JTS2300QUR

This series is titled Quits: Construction and has a frequency of Monthly. The units are Rate and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2000-12-01 and the observation end date is 2026-07-01. The popularity of this series is 13.

Series ID: WFHCOVIDFRACMATWOMEN

This series is titled Work from Home Rate: Women and has a frequency of Monthly. The units are Percent of Full Paid Working Days and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2020-05-01 and the observation end date is 2026-08-01. The popularity of this series is 6.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_WFHCOVIDFRACMATWOMEN	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.013
Method:	Least Squares	F-statistic:	0.03323
Date:	Thu, 10 Sep 2026	Prob (F-statistic):	0.856
Time:	15:17:47	Log-Likelihood:	-240.33
No. Observations:	74	AIC:	484.7
Df Residuals:	72	BIC:	489.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	32.4960	3.704	8.774	0.000	25.113	39.879
value_fred_JTS2300QUR	-0.3289	1.804	-0.182	0.856	-3.926	3.268

Omnibus:	71.083	Durbin-Watson:	0.194
Prob(Omnibus):	0.000	Jarque-Bera (JB):	449.811
Skew:	2.992	Prob(JB):	2.11e-98
Kurtosis:	13.492	Cond. No.	12.7

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

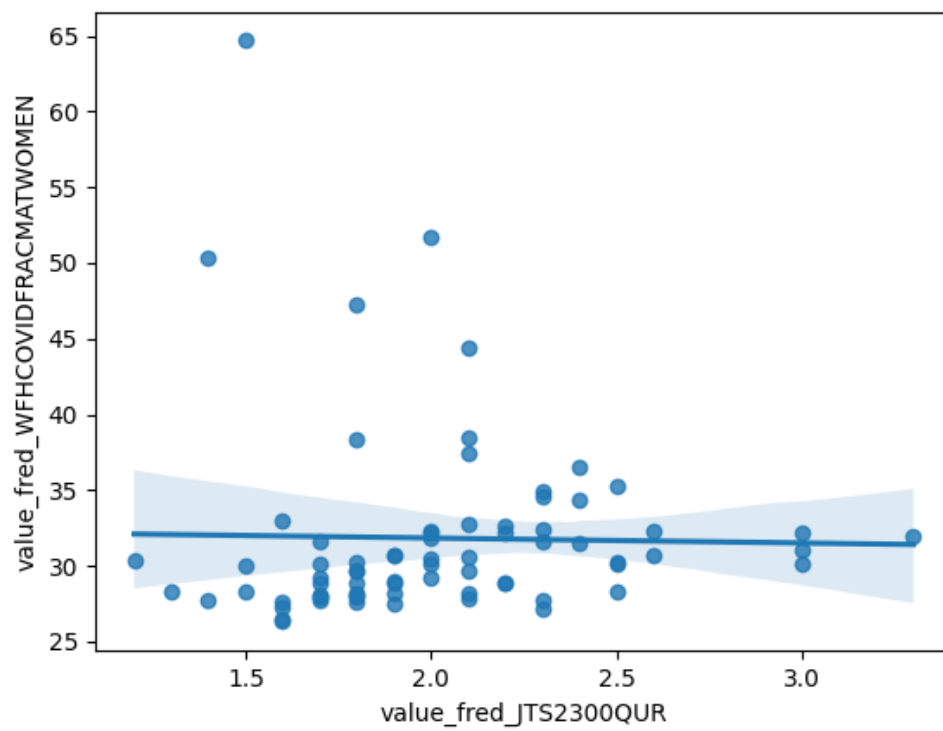


Figure 7: Raw Regression Plot for 2026-09-10

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_WFHCOVIDFRACMATWOMEN_detrended	R-squared:	0.300
Model:	OLS	Adj. R-squared:	0.290
Method:	Least Squares	F-statistic:	30.84
Date:	Thu, 10 Sep 2026	Prob (F-statistic):	4.44e-07
Time:	15:17:47	Log-Likelihood:	-205.78
No. Observations:	74	AIC:	415.6
Df Residuals:	72	BIC:	420.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-7.578e-12	0.460	-1.65e-11	1.000	-0.917	0.917
value_fred_JTS2300QUR_detrended	-7.2666	1.309	-5.553	0.000	-9.875	-4.658

Omnibus:	45.009	Durbin-Watson:	0.681
Prob(Omnibus):	0.000	Jarque-Bera (JB):	171.162
Skew:	1.828	Prob(JB):	6.80e-38
Kurtosis:	9.492	Cond. No.	2.84

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

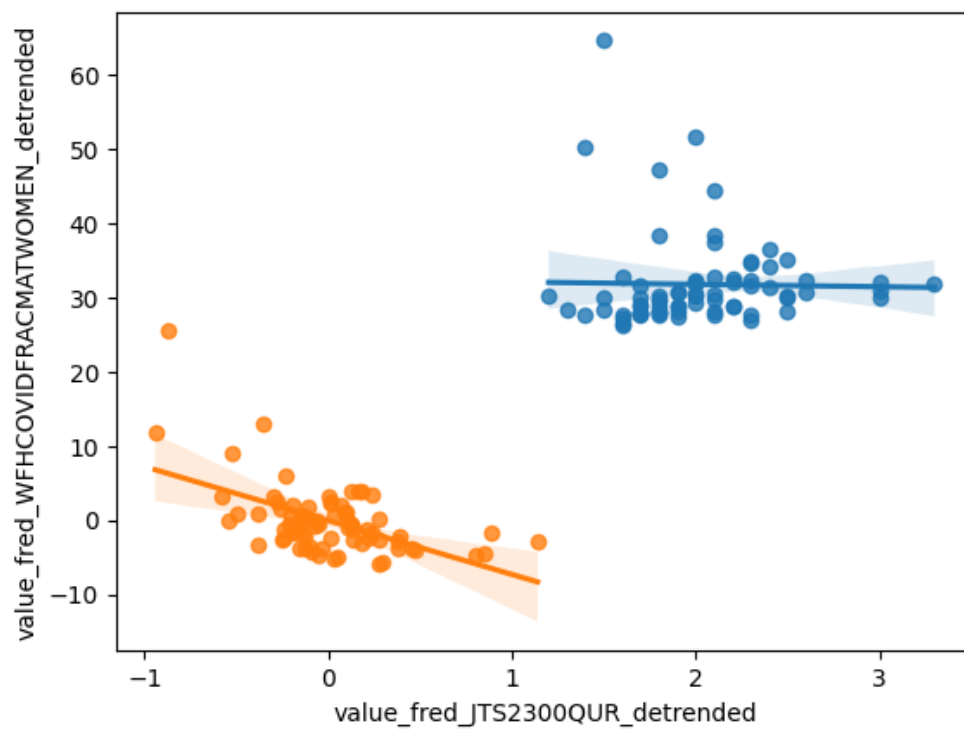


Figure 8: Detrended Regression Plot for 2026-09-10